

Design a Web Crawler - HTML Downloader

1. What is HTML Downloader?

- HTML Downloader is responsible for:
 - **Downloading web pages from the internet**
- Uses:
 - **HTTP protocol**

2. Robots Exclusion Protocol (robots.txt)

What is robots.txt?

- A **standard protocol** used by websites
- Defines:
 - Which pages crawlers are **allowed or disallowed** to access

Important Rule

Before crawling any website:

- Crawler must:
 - Fetch **robots.txt**
 - Follow its rules

Example (Amazon robots.txt)

None

User-agent: Googlebot

Disallow: /creatorhub/*

Disallow: /rss/people/*/reviews

Disallow: /gp/pdp/rss/*/reviews

Disallow: /gp/cdp/member-reviews/

Disallow: /gp/aw/cr/

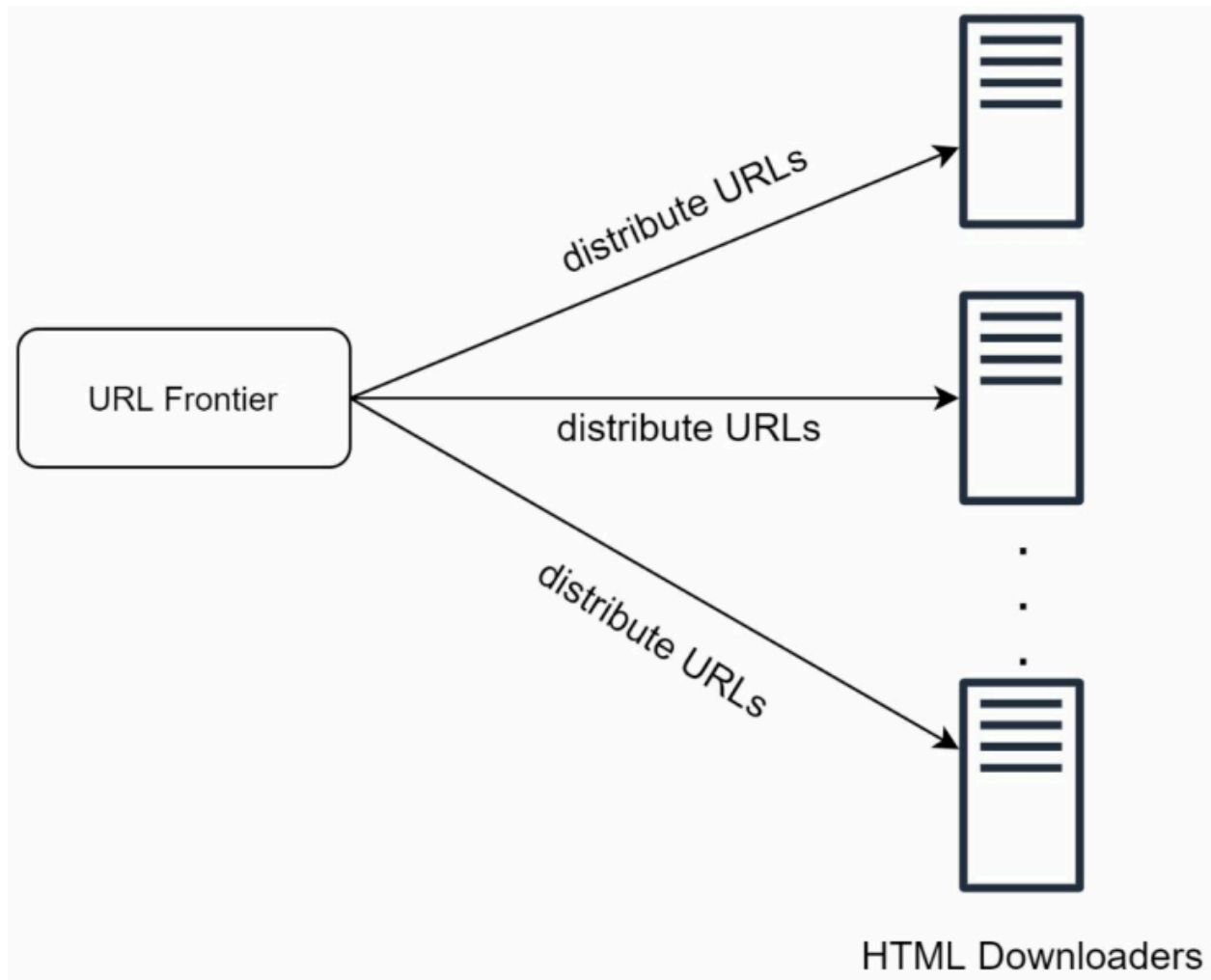
Optimization

- Avoid repeated downloads of robots.txt

Solution:

- Cache robots.txt
- Periodically refresh using background jobs

3. Performance Optimization



3.1 Distributed Crawl

Concept

- Distribute crawling workload across:
 - Multiple servers
 - Multiple threads

How It Works

- URL space is **partitioned**
- Each server:
 - Handles a subset of URLs

Benefits

- High throughput
- Parallel processing
- Better scalability

3.2 Cache DNS Resolver

Problem

- DNS resolution is **slow and blocking**

Delay:

- 10ms to 200ms per request

Issue

- Threads get blocked while waiting for DNS response

Solution

- Maintain **DNS cache**

How It Works

- Store:
 - Domain → IP mapping
- Periodically updated using:
 - Cron jobs

Benefits

- Reduced latency
- Faster crawling
- Less blocking

3.3 Locality

Concept

- Place crawl servers **closer to target websites**

Applies To

- Crawl servers
- Cache
- Queue
- Storage

Benefits

- Faster downloads
- Reduced network latency

3.4 Short Timeout

Problem

- Some servers:
 - Respond slowly
 - Do not respond

Issue

- Long wait time wastes resources

Solution

- Define **maximum timeout**

Behavior

- If server does not respond within time:
 - Stop request
 - Move to next URL

Benefits

- Efficient resource usage
- Avoid blocking

4. Key Takeaway

HTML Downloader:

- Fetches web pages using HTTP
- Respects **robots.txt** rules
- Optimized using:
 - Distributed crawling
 - DNS caching
 - Locality
 - Timeout control